Kalpā Impact

ARTHA GLOBAL

UNDP

Digital Transformation Support to UNDP Country Offices in Asia Pacific

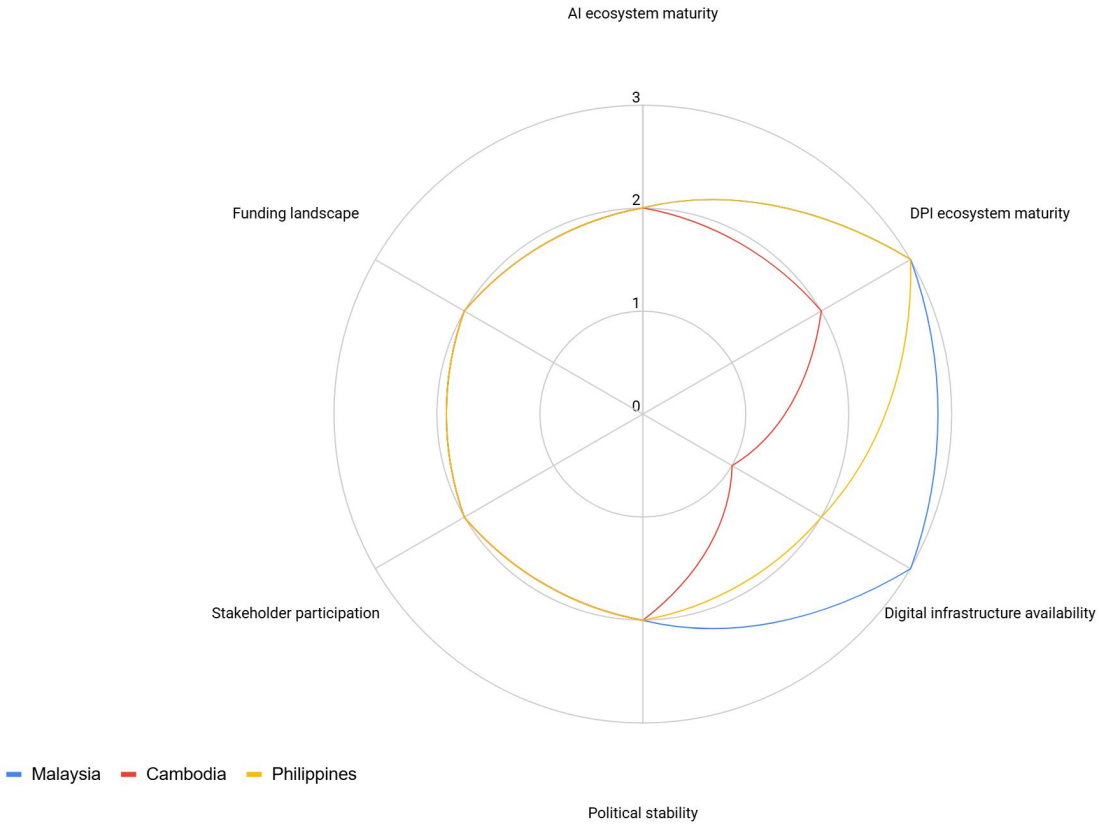
Deliverable 1: Scoping Report

Philippines, Bangladesh, Nepal, Cambodia, Malaysia

Mapping of countries across six parameters

#	Country/ Parameters	AI ecosystem maturity	DPI ecosystem maturity	Digital infrastructure availability	Political stability	Stakeholder participation	Funding landscape
1	Malaysia	Early Success	Maturing	Maturing	Early Success	Early Success	Early Success
2	Cambodia	Early Success	Early Success	Open to adopt	Early Success	Early Success	Early Success
3	Philippines	Early Success	Maturing	Early Success	Early Success	Early Success	Early Success
4	Nepal						
5	Bangladesh						

Country maturity assessment: AI and DPI ecosystem



For representation purpose only

Framework for categorisation of parameters (1/2)

#	Parameter/Category	Greenfield	Open to Adopt	Early Success	Maturing	Role Model
1	**AI ecosystem maturity**	No mention of intent to draft a Central AI policy or legislation	No national AI strategy OR ad-hoc initiatives only. Government showed intent to adopt AI policy.	National AI strategy published; governance frameworks defined; initial public-sector pilots underway	Strategy implemented with measurable targets; dedicated budget; regulatory enforcement in practice	Dedicated investment for capacity building and R&D for AI, education for public-sector workers with promotion of private actors through investment
2	**DPI ecosystem maturity**	Plan/Pilot implementation DPI Asset, or no plans yet	1 foundational DPI operational; limited operability	At least 2 foundational DPIs operational and interoperable	3 foundational DPIs operational and interoperable; 2 foundational DPIs have at least 50% TAM adoption	Foundational DPIs operational, cross-sectoral, and interoperable with widely used use-cases with an Early Succession rate for citizen-linked DPIs to be >70%
3	**Digital infrastructure availability**	Electricity or internet penetration <70%; backbone infrastructure incomplete, with no reliable evidence of any expansion underway	Electricity or internet penetration <70%; backbone infrastructure incomplete, with expansion underway	Electricity & internet penetration - >80% ; basic cloud/data hosting exists	Electricity & internet penetration - >95% ; compute-capable data centres; cloud infrastructure	~100% Electricity & Internet penetration, compute and cloud sovereignty, provides infrastructural guidance to other countries

Framework for categorisation of parameters (2/2)

#	Parameter/Category	Greenfield	Open to Adopt	Early Success	Maturing	Role Model
4	**Political stability**	Scores for governance indicators (WB) - Political stability - 0-20 percentile	Scores for governance indicators (WB) - Political stability - 20+ - 40 percentile	Scores for governance indicators (WB) - Political stability - 40+ - 60 percentile	Scores for governance indicators (WB) - Political stability - 60+ - 80 percentile	Scores for governance indicators (WB) - Political stability - 80+ - 100 percentile
5	**Stakeholder participation**	No formal AI/DPI initiative; no designated lead agency for digital transformation/AI/DPI	Single ministry/agency leading DT; limited cross-sector coordination	Inter-ministerial coordination mechanism established; structured consultations with private sector, academia, and MDBs; stakeholder input documented in strategy development	Institutionalised multi-stakeholder platforms (advisory councils, working groups); regulatory sandboxes; recurring engagement cycles; civil society participation in governance/ethics discussion	Continuous co-creation model across government, private sector, academia, and civil society; transparent decision-making
6	**Funding landscape**	No dedicated AI/DPI budget; no identifiable funding line	Ministry or agency identified; limited annual budget allocation OR externally supported funding - primarily for strategy/pilot	Ministry responsible for AI/DPI development and upkeep provided with dedicated budget OR Recurrent investment through partnership with MDBs	Sustained multi-year public investment; budget execution rates reported; domestic co-financing mechanisms	Long-term strategic or sovereign funding mechanisms; significant R&D allocation, strong domestic VC/private capital ecosystem

Scoping Report

1. Malaysia
2. Cambodia
3. Philippines
4. Nepal
5. Bangladesh

Scoping Report

1. **Malaysia**
2. Cambodia
3. Philippines
4. Nepal
5. Bangladesh

Malaysia (1/4)

Digital Public Infrastructure (DPI) — Maturing

Digital ID

1
- Implementation status: Implemented
- Program details: MyDigital ID I 2024 I RM80mn allocation
- Implementation agency: My Digital ID Sdn Bhd, Ministry of Digital
- Early Succession: 8.7 million users (~33%)

Digital payments system

2
- Implementation status: Implemented
- Program details: Real-time Retail Payments Platform I 2018, DuitNow
- Implementation agency: PayNet
- Early Succession: 5.3 bn transactions in 2024, RM5.4T transaction value

Data exchange system

3
- Implementation status: Implemented
- Program details: Malaysian Government Central Data Exchange (MyGDX) I 2018
- Implementation agency: MAMPU (Malaysian Administration Modernization and Management Planning Unit)
- Early Succession:657 users, 35 million transactions

Other DPI assets

4
- Government service delivery portal: MyGovernment managed by National Digital Department

Artificial Intelligence(AI) Ecosystem — Early Success

National AI policy/strategy

1
- Implementation status: Implemented
- Policy details: Malaysia's Artificial Intelligence Roadmap 2021-2025 I 2022
- Implementation agency: Malaysian Ministry of Science, Technology, and Innovation (MOSTI)

AI governance and framework regulations

2
- Implementation status: Implemented
- Policy details: 2 guidelines: 1. The national guidelines on AI governance and ethics; 2. ASEAN guide on AI governance and ethics
- Implementation agency: MOSTI

Data protection and privacy legislation

3
- Implementation status: Implemented
- Legislation details: Personal Data Protection (Amendment) Act 2024
- Implementation agency: Department of Personal Data Protection (JPDP)

Government AI initiatives and projects

4
NAIO is promoting the practical application of AI across sectors Healthcare, Transportation, Agriculture, Public Services, Education and MSME (More details)

National digital transformation strategy: Malaysia Digital Economy Blueprint 2021-30 I RM70 bn investment in digitalisation

Malaysia (2/4)

Availability of infrastructure — Maturing

1 Access to electricity

100% (UNESCO Readiness assessment methodology)

2 Internet penetration rate

97.7% (UNESCO Readiness assessment methodology)

3 Data centre and cloud infrastructure

Hosts 32 operational data centres with 19 under construction (UNESCO Readiness assessment methodology)

4 Access to devices

99.5% (Department of Statistics, Malaysia)

5 Digital inclusion gaps

Demographic divide: (Gender, socioeconomic) - None
Geographic: (Urban/rural)-None
Nearly equal access for men and women, urban and rural areas

Political stability and institutional capacity — Early Success

1 Ranking in the Democracy Index 2024

44 out of 167 countries (categorised as flawed democracy)

2 Improvement in political stability (2014 to 2024)

Political stability improved from 2014 to 2024 by ~5 percentage points (67.4 to 72.47) - World Bank Group

3 National election cycle

Previous and Upcoming Month and Year

5 years cycle I Previous: Nov 2022 I Upcoming: Nov-Dec 2027

4 Institutional capacity for policy implementation

Malaysia implemented various strategies and programs to develop digital competencies within the public sector. Key initiatives include:

A. Malaysia Digital Economy Blueprint (MDEB)
B. Public Sector ICT Strategic Plan
C. AI Early Succession in the public sector
D. National Talent Framework
E. Partnerships with the private sector to provide targeted digital skills training

Malaysia (3/4)

Mapping of key ecosystem actors	Maturing	Funding landscape	Early Success

#	Category	Organisation(s)	Role(Decision-maker/implementer/policy influencer)	Major initiatives	Budget/funding/investment
1	Government actors (National-level)	MOSTI, Ministry of Digital (MD), JDN (MAMPU), NAIO, MDEC, MCMC	Decision-makers & Implementers: Set national AI roadmaps, ethics guidelines, and digital infrastructure standards.	National AI Roadmap 2021–2025 (AI-RMap), National Guidelines on AI Governance and Ethics (AIGE), MyDIGITAL.	RM285M to MOSTI for AI governance; RM10M for National AI Office (NAIO) (2025 Budget).
2	Private actors	Petronas, JCorp, YTL, WISE AI, Phison, Google, Microsoft, Grab	Implementers & Influencers: Drive AI Early Succession in core sectors (energy, finance, retail) and build physical digital infrastructure.	Data centre development (YTL), Smart City initiatives, and cloud region expansions.	USD 18B+ foreign AI-related investment (2024); USD 2.2B (Microsoft) & USD 2B (Google).
3	Development partners and multilaterals	UNESCO, ITU, European Union, World Bank, UNDP	Policy Influencers & Funders: Provide technical assistance, readiness assessments, and international standard alignment.	AI Readiness Assessment Methodology (RAM), UNESCO Ethics Recommendation implementation.	Project-based funding (e.g., DFAT Australia for ITU standards project; EU for Ethics tools).
4	Academic and research institutions	UTM, UM, UPM, UKM, CAIRO (UTM), MIMOS	Policy Influencers & Implementers: Lead R&D, develop talent pipelines, and contribute technical expertise to ethics frameworks.	AI Sandbox 2024 Pilot, Faculty of AI (UTM), National AI Research Consortium.	RM50M for AI education in research universities; RM600M for general R&D and AI (2024/2025).

Challenges

Stakeholder Concerns and Expectations

Consultations revealed several recurring concerns:

- Weak regulatory and institutional support
- Poor data governance and sharing
- Fragmented coordination
- Low public awareness and trust
- Limited localization and ownership
- Skills gaps and talent drain
- Infrastructure challenges

Stakeholders also outlined expectations:

Government and Public Sector: Seek improved public service delivery, ethical guidelines, robust governance, data protection, and continuous upskilling for civil servants. AI adoption should be inclusive, transparent, and aligned with national priorities.

Industry: Expects innovation-friendly policies, co-investment opportunities, financial incentives, IP protection, and collaboration with regulators and academia. Clearly defined roles and guidance within the AI ecosystem are essential for navigating compliance, fostering ethical leadership, and ensuring accountability.

Academia: Calls for secure research facilities, high-performance computing, and access to data. They advocate for R&D in fairness and transparency, and development of AI ethics curricula.

Society: Demands transparency, awareness campaigns, public participation in policymaking, strong data rights, and equitable AI outcomes. Citizen-centric applications that ensure fairness and uphold public trust are vital.

Recommendations for an AI-Integrated Innovation Ecosystem

Part 3 outlines 19 policy recommendations across regulation, institutions, capacity-building, and infrastructure. Key priorities include:

- Establishing multi-stakeholder governance to update regulations and align with UNESCO's Ethics Recommendation
- Developing dynamic, inclusive policy frameworks and sector-specific AI regulations
- Enforcing ethical practices through compliance audits and collaboration among regulators, industry, and civil society
- Aligning AI Curriculum with Innovation and Market Relevance
- Building a strong national data governance framework covering sharing, privacy, and ownership

In the public sector, institutional capacity must be strengthened through digital and AI literacy training, creation of specialized roles, promotion of open science, and innovation hubs. Localized AI solutions and datasets should integrate indigenous knowledge.

Workforce development is crucial. Sector-specific training, expanded micro-credentials, and early education in AI ethics and digital literacy are needed. Talent attraction and retention require competitive policies, diversity promotion, and support for AI champions and trainers.

Infrastructure expansion, particularly in underserved areas, and investment in green AI (e.g., energy-efficient data centres) will support inclusive, responsible development. Stronger collaborations across sectors and real-world projects will sustain the talent pipeline and achieve Malaysia's long-term AI goals. Stakeholders' concerns, expectations, and recommendations are summarized in Table 1 below.

- Limited technical expertise and resources.
- Dependency on imported AI solutions, leading to fewe
- The absence of a dedicated regulatory framework.
- The need to harmonize AI governance with existing se

The Malaysian National AI Office (NAIO), established on 28
Malaysia, seeks to accelerate AI adoption, foster innovation,
of artificial intelligence. It aims to enhance Malaysia's con

Challenges

- **Enforceable Legislation**
  Introducing legal mandates to ensure consistent publication of high-impact datasets across government agencies.
- **Data Quality Enhancement**
  Improving standardization, completeness, and regular updates of datasets to maximize their utility.
- **Public Awareness and Engagement**
  Conducting education and capacity-building initiatives to promote the effective use of open data.
- **Private Sector Collaboration**
  Encouraging greater private sector involvement to drive innovation and address public benefit challenges.

Malaysia (4/4)

Emerging opportunities

Public service AI integration

1 → Expanding the "AI at Work" initiative to enhance efficiency across 10 key socio-economic sectors

Rapid startup growth

2 → Utilising the AI Sandbox to increase the number of AI technology companies from 284 to 900 by 2026

Regional AI leadership

3 → Leveraging the NAIO and the 2024 AIGE guidelines to position Malaysia as a regional hub for ethical AI

Potential risks

Governance and regulatory risks

1 → Legal infrastructure requires significant updates, particularly in data protection and enforceable ethical standards

Loss of cultural and linguistic heritage

2 → A lack of explicit policies addressing AI's impact on cultural preservation and indigenous languages could lead to a loss of national identity

Dependence on foreign Tech

3 → There is a high cost and potential over-dependence on foreign AI vendors which can limit homegrown innovation and sovereign control over AI solutions

Partnership and engagement opportunities for UNDP

Green AI advocacy

1 → Working with the Ministry of Digital to establish monitoring systems and policies for sustainable data centers

Public sector digital literacy

2 → Support the Department of National Digital (JDN) in implementing the "Public Sector AI Adaptation Guidelines". UNDP can help design training modules for civil servants

Strategic innovation-AI Sandbox

3 → Collaborate with the AI Sandbox initiative to provide grants and mentorship for startups focusing on AI for Good

Operationalising ethical impact assessments

4 → Partner with the NAIO and MOSTI to pilot UNESCO's EIA tools Ensure that public sector AI Early Succession remains human-centric, transparent, and accountable.

Scoping Report

1. Malaysia
2. **Cambodia**
3. Philippines
4. Nepal
5. Bangladesh

Cambodia (1/4)

Digital Public Infrastructure (DPI) — Early Success

Digital ID

1
- Implementation status: Implemented
- Program details: CamDigiKey | 2019
- Implementation agency: Ministry of Economy and Finance
- Adoption: 86k Users

Digital payments system

2
- Implementation status: Implemented
- Program details: Bakong I 2020
- Implementation agency: National Bank of Cambodia
- Adoption: 60% adult population as of 2021

Data exchange system

3
- Implementation status: Implemented
- Program details: CamDX I 2020
- Implementation agency: Ministry of Economy and Finance

Other DPI assets

4
a. Verify.gov Cambodia, used for government document verification in Cambodia.

Artificial Intelligence(AI) Ecosystem — Early Success

National AI policy/strategy

1
- Implementation status: Drafted
- Policy details: National AI Strategy I 2024
- Implementation agency: Ministry of Post and Telecommunications

AI governance and framework regulations

2
- Implementation status: Drafted
- Policy details: National AI Governance Framework | 2024
- Implementation agency: Ministry of Post and Telecommunications

Data protection and privacy legislation

3
- Implementation status: Drafted
- Legislation details: Law on Personal Data Protection I 2025
- Implementation agency: Ministry of Post and Telecommunications

Government AI initiatives and projects

4
- NEC and the Cambodian Mine Action Center Successfully Predict Landmine
- National Research Center on AI for Education inaugarated (Nov 2025)
- Cambodia and Singapore Partner to Develop Khmer Large Language Model

Cambodia (2/4)

Availability of infrastructure

Open to Adopt

1 Access to electricity

92.3% in 2022 (UNESCO AI Readiness Report)

2 Internet penetration rate

60%+ (ITU)

3 Data centre and cloud infrastructure

4 Commercial Data Centres out of which only 3 are considered modern (UNESCO AI Readiness Report)

4 Access to devices

High % of Mobile Phone ownership, while low penetration of CPU and GPU servers at an institutional level (Science Council)

5 Digital inclusion gaps

Gender divide more evident in rural areas for digital access

Political stability and institutional capacity

Early Success

1 Ranking in the Democracy Index 2024

Authoritarian Regime - 123/169

2 Improvement in political stability (2014 to 2024)

~10% increase in Percentage points for Political Stability index according to World Bank

3 National election cycle

Previous and Upcoming Month and Year

Previous cycle in 2023, next one in 2028

4 Institutional capacity for policy implementation

Project	Lead Agency	Target Date
National Cybersecurity Law	MPTC	Mid-2026 (Ratification)
Personal Data Protection Law	MEF / MPTC	Late 2026 (Drafted)

Cambodia (3/4)

Mapping of key ecosystem actors

Early Success

Funding landscape

Early Success

Category	Organisation(s)	Role	Major Initiatives	Budget / Funding
Government (National)	**Ministry of Post & Telecom (MPTC)**; MISTI; MoEYS	**Decision-maker & Lead**	**National AI Strategy (2024-2035)**: Focused on 6 priorities including "AI for Good" and Khmer LLM development (e.g., *Sarika* voice-to-text).	Included in the **$10.1B** national budget; specific AI/Digital R&D is targeted to rise from current 0.09% GDP.
Private Actors	**Techo Startup Center**	**Implementer**	**AI-driven Public Services**: Implementation of biometric eID, liveness checks (OCR), and fraud detection in the Bakong payment system.	Part of **$10B** in registered investment projects; major focus on Fintech and logistics AI.
Development Partners	**UNESCO**; UNDP; Asian Development Bank (ADB)	**Policy Influencer / Funder**	**RAM Implementation**: Turning UNESCO's 2021 Ethics Recommendations into local laws; **$50M ADB loan** for digital infrastructure.	~$3.1B in concessional loans/SDRs provided to ADB; UNESCO provides technical expertise via "AI Ethics Experts without Borders."
Academic & Research	**Cambodia Academy of Digital Tech (CADT)**;	**Implementer / Policy Influencer**	**Research Area in AI for Education (RAIE)**: A national hub for interdisciplinary research on AI's impact on learning and pedagogical ethics.	**$27M** via STEPCam Phase II (UNESCO/MoEYS) for teacher prep; LBE Research Grants.
Civil Society	**Open Development Cambodia (ODC)**; NGO Forum	**Watchdog / Policy Influencer**	**Algorithmic Accountability**: Monitoring the impacts of AI on data privacy and the gender gap in STEM/Digital Tech.	Primarily funded by international grants; focus on transparency in public-sector AI procurement.
Philanthropic & Funding	**Konrad-Adenauer-Stiftung (KAS)**; Coca-Cola Foundation	**Funder / Capacity Builder**	**Digital Insights Series**: Funding policy-relevant research to bridge the gap between AI possibility and actual policy action.	Variable; focused on "AI Essentials for Work" training for civil servants and SMEs.

Cambodia (4/4)

Emerging opportunities

1 Early Succession of e-ID
- eID has low Early Succession rates in Cambodia, majorly availed for Business-related government service
- Can make sure marginalized populations aren't left behind

2 National AI Strategy implementation
- AI Safety with operationalizing AI Sandboxes for public health and governance

3 Infrastructure Bottlenecks
- Connectivity in terms of internet and electricity is improving constantly, GPU and CPU servers remain scarce in institutes though

Potential risks

1 Deepening Digital Divide
- Rapid 5G and AI Early Succession risk leaving rural communities and women behind.

2 Data Vulnerability & Privacy
- With the **Law on Personal Data Protection** still in the drafting/ratification stage, there is a risk of data misuse

3 Institutional Fragmentation
- Digital initiatives are currently split across MPTC, MISTI, and MEF along with separate working groups for Digital Initiatives and the National Bank of Cambodia

Partnership and engagement opportunities for UNDP

1 South-South Cooperation
- UNDP can facilitate knowledge exchange between Cambodia and other regional leaders (like Singapore or Vietnam) on DPI Implementations and institutional infrastructure bottlenecks

2 Policy Support for AI Ethics
- Building on the **UNESCO AI Readiness Assessment**, partners can provide technical expertise to help turn ethical recommendations into enforceable local laws.

3 Joint Human Capital Initiatives
- Collaborate with **CADT** and **UNESCO** on the **$27M STEPCam Phase II** to ensure the curriculum includes "Future of Work" skills and AI ethics.

4 Private Sector De-risking
- Work with the **Techo Startup Center** to provide grants or technical mentorship for "Impact Tech" startups that solve local developmental challenges using AI.

Scoping Report

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Philippines (1/4)

Digital Public Infrastructure (DPI) — Mature

Digital ID

1
- Implementation status: Implemented
- Program details: PhilSys/ PhilID | 2018 | ₱30B
- Implementation agency: Philippine Statistics Authority (PSA)
- Early Succession: 90.02 Million registered (~80% Population)

Digital payments system

2
- Implementation status: Implemented
- Program details: InstaPay, PESONet | 2017
- Implementation agency: Bangko Sentral ng Pilipinas (BSP)
- Early Succession: ₱24.745 trillion (Total transaction, end of 2025)

Data exchange system

3
- Implementation status: Implemented
- Program details: eGov PH Super App & eGovDX | 2024
- Implementation agency: DICT (Dept. of Information and Communications Technology)
- Early Succession: 5M+ downloads; 42 out of 75 NGAs integrated.

Other DPI assets

4
a. Government service delivery portal: eGov PH Portal (one-stop-shop for LGU/NGA services).
b. Verifiable credentials: PhilSys Check (digital ID authentication) and eTravel system.

Artificial Intelligence(AI) Ecosystem — Early Success

National AI policy/strategy

1
- Implementation status: Implemented (Updated)
- Policy details: National AI Strategy Roadmap 2.0 | 2024
- Implementation agency: Department of Trade and Industry (DTI)

AI governance and framework regulations

2
- Implementation status: Planned / In Progress
- Policy details: AI Regulation Act (AIRA - SB 25 / HB 6028) | 2025 | UNESCO AI Readiness Assessment
- Implementation agency: DTI / DOST / Philippine Congress

Data protection and privacy legislation

3
- Implementation status: Implemented
- Legislation details: Data Privacy Act of 2012 (RA 10173) | 2012
- Implementation agency: National Privacy Commission (NPC)

Government AI initiatives and projects

4
Details: Establishment of the Center for AI Research (CAIR); ALAM Project (machine learning optimization); and iTANONG (AI natural language querying for public info).

National digital transformation strategy: Digital transformation is included in the Philippines Development Plan 2023-28

Philippines (2/4)

Availability of infrastructure

Early Success

1 Access to electricity

~97.5% (World Bank Open Data)

2 Internet penetration rate

83.8% (Digital 2026 Report: Philippines)

3 Data centre and cloud infrastructure

28 operational data centres; with 13 upcoming facilities (GlobalNewsWire)

4 Access to devices

98.6% (Smartphone ownership %; Digital 2026 Report: Philippines)

5 Digital inclusion gaps

Geographic divide: Significant gap between Metro Manila (~68.7% penetration) and regions like BARMM (27.7%) or Zamboanga (21.2%)

Political stability and institutional capacity

Early Success

1 Ranking in the Democracy Index 2024

#51 out of 167 countries (categorised as flawed democracy)

2 Improvement in political stability (2014 to 2024)

+0.1478 points improved (-0.7138 in 2014 to -0.5661 in 2023, World Bank Political Stability Index).

3 National election cycle Previous & Upcoming Month & Year

6-year presidential cycle | Prev: May 2022 | Next: May 2028
3-year midterm cycle (Senate half, House, local) | Prev: May 2025 | Next: May 2028

4 Institutional capacity for policy implementation

Details of ongoing/pipeline initiatives:

A. Philippine Digital Infrastructure Project (PDIP): $273M World Bank-funded project to bridge the broadband gap
B. National Fiber Backbone (NFB): A multi-phase project aiming to raise fixed broadband penetration to 65% by 2026
C. Center for AI Research (CAIR): DTI-led initiative to foster AI-driven innovation and governance

Philippines (3/4)

Mapping of key ecosystem actors

Early Success

Funding landscape

Early Success

#	Category	Organisation(s)	Role (Decision-maker/ implementer/ policy influencer)	Major initiatives	Budget/funding/ investment
1	Government actors (National-level)	DICT, DTI, DOST, PSA	**Decision-makers & Implementers:** Set national roadmaps and infrastructure standards	eGovPH Superapp, PhilSys, National AI Strategy Roadmap 2.0	₱18.9 Billion (DICT 2026 Proposed Budget)
2	Private actors	GCash (Mynt), Maya, Globe, PLDT, Grab	**Implementers & Influencers:** Drive digital payments and commercial connectivity	Expansion of digital banking (Maya) and e-wallets (GCash)	$4.1 Billion (GCash valuation); ~$2B in data centers by 2026
3	Development partners and multilaterals	UNDP, World Bank, ADB, UNESCO, ILO	**Policy Influencers & Funders:** Provide technical assistance and large-scale loans	Digital-PINAS, Philippine Digital Infrastructure Project (PDIP)	$378M (World Bank/AIIB Loan); $273M PDIP
4	Academic and research institutions	UP, DLSU, ADMU, Center for AI Research (CAIR)	**Policy Influencers & Implementers:** Lead R&D and digital talent development	AI4RP (Weather Forecasting), GATES Program (Geospatial AI)	DOST-funded R&D grants; Startup Venture Fund
5	Civil society	Foundation for Media Alternatives (FMA), Bulatlat, Rights at the Core	**Policy Influencers:** Advocate for digital rights, privacy, and inclusion	"Rights-Respecting Digital Ecosystem" regional briefing	Primarily grant-funded (multilateral/philanthropic)
6	Philanthropic and funding orgs	AVPN, Joint SDG Fund, Gobi-Core PH Fund	**Funders:** Provide early-stage capital and social impact investment	AI Opportunity Fund (AVPN), Startup QC Grants	$80M+ raised by AI startups in 2024-25

Philippines(4/4)

Emerging opportunities

1 Accelerating National AI Policy Momentum

→ Release of NAISR 2.0 and updated DOST AI Roadmap reflects strong government commitment to advancing AI Early Succession

2 Expanding Digital Economy & High AI Adoption

→ Digital economy drives significant GDP contribution, with above global AI Early Succession and strong projected economic gains

3 Growing Multi-Stakeholder Engagement on AI

→ Active participation from industry, academia, and civil society supports a whole-of-nation approach to AI development

Potential risks

1 Siloed AI Policymaking & Lack of Lead Agency

→ Absence of a dedicated coordinating mechanism weakens policy coherence and implementation effectiveness

2 Persistent Infrastructure & Connectivity Gaps

→ Government R&D spending remains below global benchmarks, limiting domestic innovation and ethical AI research outputs

3 Public Trust (Cybersec & Labor Disruption Risks)

→ Cyber threats, deepfakes, and potential job displacement raise concerns over public trust and economic stability

Partnership and engagement opportunities for UNDP

1 Support Establishment of AI Agency Coordination

→ Assist in operationalizing a dedicated inter-agency (DTI, DICT, DOST, NEDA) mechanism to address siloed AI challenges

2 Increase Capacity Building and R&D Support

→ Facilitate initiatives to expand funding, strengthen industry –academia collaboration, and boost domestic AI research

3 Strengthen Ethical AI Policy Integration

→ Support contextualizing AI ethics within Filipino socio-cultural frameworks and promote responsible AI awareness

4 Enhance Data Governance & Cybersec Frameworks

→ Support robust data governance frameworks, cross-sector data sharing mechanisms and AI cybersecurity risk mitigation

Parameters - scaling according to the features

Parameter	Early Success	Mature	Exemplar
AI Ecosystem	No Nationwide Strategy	National Strategy drafted - with multiple stakeholders	National Strategy Published and progress made on accomplishing goals
DPI Ecosystem	1 DPI implemented, others planned	2 DPIs implemented, others planned/being implemented	3 DPIs implemented and Early Successed by 70%+ of the TAM
Digital Infrastructure	Physical infrastructure still being laid out as part of increasing connectivity	Connectivity to Electricity and Internet above 90%	Connectivity to E&I above 90% and several Compute capable data centres for low latency compute
Political Stability	World Bank scores on Political Stability and Democracy Index are low and stable/trending downwards	World Bank scores on Political Stability and Democracy Index are low/medium and trending upwards	World Bank scores on Political Stability and Democracy Index are high and stable/trending upwards
Stakeholder Engagement	Specific ministry/agency/working group present in government for digital transformation	Ministry responsible for DT has engaged MDBs, ADBs, or UN organizations	Ministry responsible for DT actively capacitates small NGOs, startups, PIOs etc. while engaging larger Multilateral Orgs
Funding Landscape	No explicitly mentioned budget allocated to ministries responsible for DPI/National AI development	Ministry responsible for AI/DPI development and upkeep provided with adequate funding with minimal cases of corruption reported OR Recurrent Investment through partnership with MDBs and ADBs	Budget explicitly allocated for the development and upkeep of AI/DPI infrastructure, with budget spent % reported on as well

Country Name (1/4)

National digital transformation strategy: "Program name, Year, IA, Budget allocation"

Digital Public Infrastructure (DPI) — XX

Digital ID

1
- Implementation status: Planned/Pilot/Implemented
- Program details: Name I Year I Budget
- Implementation agency:
- Early Succession:

Digital payments system

2
- Implementation status: Planned/Pilot/Implemented
- Program details: Name I Year I Budget
- Implementation agency:
- Early Succession:

Data exchange system

3
- Implementation status: Planned/Pilot/Implemented
- Program details: Name I Year I Budget
- Implementation agency:
- Early Succession:

Other DPI assets

4
a. Government service delivery portal
b. Verifiable credentials or any other DPI Asset

Artificial Intelligence(AI) Ecosystem — XX

National AI policy/strategy

1
- Implementation status: Planned/Pilot/Implemented
- Policy details: Name I Year I Budget
- Implementation agency:

AI governance and framework regulations

2
- Implementation status: Planned/Pilot/Implemented
- Policy details:
- Implementation agency:

Data protection and privacy legislation

3
- Implementation status: Planned/Pilot/Implemented
- Legislation details: Name I Year
- Implementation agency:

Government AI initiatives and projects

4
a. Details of ongoing/pipeline initiatives

Country Name (2/4)

Availability of infrastructure | XX

1. **Access to electricity**
   % (Source)

2. **Internet penetration rate**
   % (Source)

3. **Data centre and cloud infrastructure**
   No. of data centres, investment in cloud infra (Source)

4. **Access to devices**
   % (Source)

5. **Digital inclusion gaps**
   Demographic divide: (Gender, socioeconomic)
   Geographic: (Urban/rural)

Political stability and institutional capacity | XX

1. **Ranking in the Democracy Index 2024**
   #

2. **Improvement in political stability (2014 to 2024)**
   Percentage points

3. **National election cycle**
   Previous and Upcoming Month and Year
   # I Previous elections I Upcoming details

4. **Institutional capacity for policy implementation**
   Details of ongoing/pipeline initiatives:
   A.
   B.
   C.

Country Name (3/4)

Mapping of key ecosystem actors	XX	Funding landscape	XX

#	Category	Organisation(s)	Role (Decision-maker/implementer/policy influencer)	Major initiatives	Budget/funding /investment
1	Government actors (National-level)				
2	Private actors				
3	Development partners and multilaterals				
4	Academic and research institutions				
5	Civil society				
6	Philanthropic and funding orgs				

Country Name (4/4)

Emerging opportunities

1. **ABC**
   → Description
2. **ABC**
   → Description
3. **ABC**
   → Description

Potential risks

1. **ABC**
   → Description
2. **ABC**
   → Description
3. **ABC**
   → Description

Partnership and engagement opportunities for UNDP

1. **ABC**
   → Description
2. **ABC**
   → Description
3. **ABC**
   → Description
4. **ABC**
   → Description

Scoping Report

1. Malaysia
2. Cambodia
3. Philippines
4. **Nepal**
5. Bangladesh

Nepal (1/4)

Digital Public Infrastructure (DPI) — XX

Digital ID

1
- Implementation status: Implemented
- Program details: Rastriya Parichaya Patra; National Identity; Launched in 2018/19
- Implementation agency: Department of National ID and Civil Registration (DoNIDCR)
- Early Success: 57% of the population has registered

Digital payments system

2
- Implementation status: Implemented
- Program details: National Payments Interface (NPI); Launched in 2024
- Implementation agency: Nepal Rastra Bank (NRB)
- Early Success: NCHL processed 186 million transactions, covering 17% of total market volume but 77% of value. Also ~90% of government expenses are digital transactions

Data exchange system

3
- Implementation status: Implemented
- Program details: Government Integrated Data Center (GIDC)
- Implementation agency: Ministry of Communication and Information Technology (MoCIT)
- Early Succession: Has improved the delivery of e-governance

Other DPI assets

4
Nagarik App: Launched in 2021, it currently offers access to 61 public services through a single e-governance platform.
SUSASAN: Has developed digital tools that have been adopted by multiple municipalities in Nepal; a portal that provides data and enables citizens to give feedback to local government

Artificial Intelligence(AI) Ecosystem — XX

National AI policy/strategy

1
- Implementation status: Implemented
- Policy details: National AI Policy, 2082 (2025)
- Implementation agency: Ministry of Communication and Information Technology (MoCIT)

AI governance and framework regulations

2
- Implementation status: Implemented
- Policy details: National AI Policy, 2082 (2025)
- Implementation agency: Ministry of Communication and Information Technology (MoCIT)

Data protection and privacy legislation

3
- Implementation status: Implemented
- Legislation details: National Cyber Security Policy, 2023
- Implementation agency: Council of Ministers, Nepal

Government AI initiatives and projects

4
Lalitpur Metropolitan City (LMC): AI-based intelligent traffic light system deployed in major junctions to adjust traffic signals based on real time data
SITA: An AI platform that rapidly analyses national datasets and generates detailed insights to help make real time decisions at national level.

National digital transformation strategy: Digital Nepal Framework 2.0 and IT Decade Vision

Nepal (2/4)

Availability of infrastructure

XX

1 Access to electricity

94.0% in 2023 (World Bank)

2 Internet penetration rate

56% (World Bank, as of 2023)

3 Data centre and cloud infrastructure

7 data centres (Internet Society: Pulse)

4 Access to devices

95.5% (National Statistics Office)

5 Digital inclusion gaps

Higher internet access rates: Urban areas (65%) Rural regions (22%) Only 53% have 4G coverage, 5% have fixed broadband and just 56% have used the internet recently (IT Infrastructure Report)

Political stability and institutional capacity

XX

1 Ranking in the Democracy Index 2024

Hybrid Regime: 4.6/10

2 Improvement in political stability (2014 to 2024)

10.87% increase in Percentage points for Political Stability index (47.94 in 2014 and 58.81 in 2024) (World Bank)

3 National election cycle

Previous and Upcoming Month and Year

November 2022 | March 2026

4 Institutional capacity for policy implementation

Participating in the 50-in-5 initiative

Nepal (3/4)

Mapping of key ecosystem actors XX

Funding landscape XX

Category	Organisation(s)	Role	Major initiatives	Budget/funding/investment
Government actors (National-level)	Ministry of Communication and Information Technology, Nepal e-governance board, Fonepay Payment service Ltd, Nepal Telecommunication Authority, Press Council Nepal, Office of the Controller of Certification	Decision-makers & Implementers	Fonepay has collaborated with NPCI to facilitate UPI payments in Nepal	
Private actors	Leapfrog, F1Soft International, Verisk Nepal, CloudFactory Nepal, AI Automation Nepal, Fusemachines Nepal	Implementers		
Development partners and multilaterals	World Bank, ADB, UNDP, Global Environment Facility, Global Climate Fund	Policy Influencers & Funders		-In 2021, UNDP Nepal implemented development programmes across Nepal with a total expenditure of $20.4 million -The World Bank and the ADB jointly co-financE the Nepal Digital Transformation Project contributing $50 million and $ 40 million respectively
Academic and research institutions	Open Knowledge Nepal	Policy Influencers		
Civil society	Digital Rights Nepal	Policy Influencers and Implementers		
Philanthropic and funding orgs	NPCI International Payments Limited (NIPL), AI Association Nepal	Funders		

Nepal (4/4)

Emerging opportunities

1. **Nepal Digital Transformation Project**
   World Bank approved a $50m loan that aims to strengthen Nepal's digital public infrastructure and improve the delivery and use of inclusive, high-impact digital government services

2. **Broadband and Data Infrastructure Expansion**
   IFC, Standard Chartered Bank Nepal Limited, and WorldLink Communications Ltd. partner on $29 million project to expand connectivity, create jobs, and drive economic growth

3. **Nepal is not part of the 50-in-5 initiative**
   Given Nepal's focus on digital infrastructure expansion, there is scope for the country to become a part of the 50-in-5 initiative

Potential risks

1. **GIDC lacks resources which results in outages/breaches**
   Government Integrated Data Center (GIDC) is understaffed and lacks resources including technical infrastructure which led to outages and data loss across 10–15 agencies in 2023

2. **Weak Systems and Limited Uptake**
   DPI understanding remains low beyond central ministries, limiting sectoral uptake. Nepal scores 0.439 on the GovTech Index, reflecting weak systems and skills across government

3. **Data Protection Landscape**
   The 2024 Draft Cyber Bill has been criticised for potentially enabling government surveillance and censorship - especially through a proposed national internet gateway - while lacking strong privacy safeguards and adequate public consultation

Partnership and engagement opportunities for UNDP

1. **Strengthening of GDIC**
   UNDP could support the institutional strengthening of the Government Integrated Data Center (GIDC) by helping design operational standards, cybersecurity protocols, and workforce capacity programmes for digital infrastructure management

2. **Focused Capacity-Building Across High-Priority Ministries**
   Since DPI understanding remains concentrated within central ministries, UNDP could run capacity-building programmes for specific ministries to increase adoption of digital systems

3. **Supporting Rights-Based Data Governance**
   With concerns around the 2024 Draft Cyber Bill, UNDP could facilitate multi-stakeholder dialogue and technical assistance on data protection and cybersecurity legislation, ensuring alignment with international norms

Scoping Report

1. Malaysia
2. Cambodia
3. Philippines
4. Nepal
5. **Bangladesh**

Bangladesh (1/4)

Digital Public Infrastructure (DPI) | XX

Digital ID

1
- Implementation status: Implemented
- Program details: National ID I 2016 I Tk 70.63 crore (for 2026-27)
- Implementation agency: Ministry of Home Affairs
- Early Succession: Youth aged 16 or above are now eligible for NID cards and services, as of February 2026

Digital payments system

2
- Implementation status: Implemented
- Program details: National Payment Switch Bangladesh I 2012 I
- Implementation agency: Bangladesh Bank
- Early Succession: Bank-to-MFS transfers fees set at Tk 1.5 per Tk 1,000; MFS-to-bank transfers set at Tk 8.5 per Tk 1,000; 2025

Data exchange system

3
- Implementation status: Implemented
- Program details: National E-Service Bus I 2019
- Implementation agency: Bangladesh National Enterprise Architecture (BNEA)
- Early Succession: Being used by 15 Bangladeshi government agencies, as of August 2025

Other DPI assets

4
Doptor: Currently offers 10+ services and 40+ applications to about 14k offices and 140k users with data on the entirety of Bangladesh to native and other populations

Artificial Intelligence(AI) Ecosystem | XX

National AI policy/strategy

1
- Implementation status: Drafted
- Policy details: National AI Policy Bangladesh 2026-30 (Drafts V1.1 and V2.0)
- Implementation agency: The Ministry of Posts, Telecommunications and Information Technology

AI governance and framework regulations

2
- Implementation status: Drafted
  Policy details: National AI Policy Bangladesh 2026-30 (Drafts V1.1 and V2.0)
- Implementation agency: The Ministry of Posts, Telecommunications and Information Technology

Data protection and privacy legislation

3
- Implementation status: Implemented
- Legislation details: The Personal Data Protection Ordinance and the Cyber Security Ordinance | 2025
- Implementation agency: The Ministry of Posts, Telecommunications and Information Technology

Government AI initiatives and projects

4
Kagoj AI: Bangladesh's first Artificial Intelligence platform for the Bangla language uses a new *Julai* font)

National digital transformation strategy: Smart Bangladesh Vision 2041

Bangladesh (2/4)

Availability of infrastructure | XX

1 Access to electricity

99.5% in 2023 (World Bank)

2 Internet penetration rate

45% (World Bank, as of 2023)

3 Data centre and cloud infrastructure

36 data centres (Internet Society: Pulse)

4 Access to devices

98.9% (Bangladesh Bureau of Statistics)

5 Digital inclusion gaps

Demographic divide: (Gender, socioeconomic) 0.767%, **Urban areas**: 71.3% of men and 62.4% of women; **Rural areas**: 36.6% of men and 23% of women (use of internet) **Geographic**: (Urban/rural) 66.8% vs 29.7% (UNESCO Readiness Assessment Report)

Political stability and institutional capacity | XX

1 Ranking in the Democracy Index 2024

Hybrid Regime: 100/167 (EIU)

2 Improvement in political stability (2014 to 2024)

7.04% decrease in Percentage points for Political Stability index (51.48 in 2014 and 40.90 in 2024) (World Bank)

3 National election cycle

Previous and Upcoming Month and Year

January 2024 | February 2026

4 Institutional capacity for policy implementation

- Building a phygital network of over 9000+ data centres
- Interbank payment system to be developed by 2027
- A three phased ICT Division Roadmap (2025-30)
- EkShop: a rural assisted, government e-commerce platform that acts as an " e-commerce infrastructure backbone" for Bangladesh.

Bangladesh (3/4)

Mapping of key ecosystem actors XX

Funding landscape XX

Category	Organisation(s)	Role	Major initiatives	Budget/funding/investment
Govt actors (National-level)	MoPTICT, Bangladesh Bank, Election Commision Porichoy	Decision-makers & Implementers	A2i (Aspire to Innovate), Smart Bangladesh, National ID	The government has prioritized digitalization, allocating ~$200 million to the ICT Division and $11 million to support digital entrepreneurship in fiscal year (FY) 2024/25
Private actors	bKash, Banglalink, Intellier	Implementers	bHive by Bkash, MyBL by Banglalink, Nidle by Intellier	
Development partners and multilaterals	World Bank, ADB	Policy Influencers & Funders	Digital Service Transformation for Access and Resilience, Enhancing Digital Government & Economy Project (EDGE), (World Bank) Promoting Artificial Intelligence-Enhanced Distribution Grids for Sustainability and Resilience (ADB)	
Academic and research institutions	AI Forum Bangladesh, BIGD, CPD	Policy Influencers	Hackathons, A2i (Aspire to Innovate), DigiDev Initiative	
Civil society	Bangladesh NGOs Network for Radio and Communication (BNNRC), Digitally Right	Policy Influencers and Implementers	Bangladesh Tech Policy Fellowship, promoting tech governance	
Philanthropic and funding orgs	Korea International Cooperation Agency (KOICA)	Funders	Fostering Smart & Eco-Friendly Ship Recycling Experts through Physical AI in Bangladesh, Fostering Innovative Technology Experts with a Focus on AI in Bangladesh, Establishment of Bangladesh National Intelligent Cybersecurity Center and AI-Integrated Security Capacity Building	KOICA grants $96m to boost Bangladesh's future in advanced tech, AI, and sustainable development

Bangladesh (4/4)

Emerging opportunities

1. **Smart Bangladesh 2041**
   - Smart Bangladesh focuses on expanding citizens' digital skills and access to services, digitizing government operations and service delivery, applying digital technologies across sectors such and promoting technology-driven economic growth
2. **Interbank payment system**
   - Description: From September 2025, the Bangladesh bank aims to develop an interoperable payment system with the Gates Foundation, built on Mojaloop-based IIPS by July 2027
3. **Phygital Public infrastructure for Bangladesh**
   - Encompassing all 3 layers of DPI, Phygital Public Infrastructure also aims to bridge the gaps between marginalised groups and public services. These include aspiring youth, women; both urban and rural

Potential risks

1. **Weak Cybersecurity**
   - Currently prone to a high volume of cyberattacks
   - Severe shortage of cybersecurity professionals
   - The NID breach of 2023 exposed about 50 million citizen data
2. **Lack of interoperability**
   - Bangladesh's data remains fragmented and siloed
   - Systems lack secure interoperability
   - Isolation of systems has also led to high operational costs
3. **Limited infrastructure and literacy**
   - Low digital literacy rate
   - Limited infrastructure and last mile connectivity issues

Partnership and engagement opportunities for UNDP

1. **Digital Skills and Capacity Development**
   Support large-scale digital literacy and workforce upskilling initiatives under programs such as Citizen Upskilling and the Digital Leadership Academy
2. **Digital Governance and Public Service Transformation**
   Support could focus on designing interoperable digital platforms, improving public service delivery, and strengthening regulatory frameworks for data governance, privacy, and cybersecurity.
3. **Digital Inclusion and Bridging the Digital Divide**
   Design programs that ensure women, rural communities, persons with disabilities, and other marginalized groups benefit from digital transformation. This could involve supporting affordable connectivity, inclusive service design, and targeted digital literacy programs.